ECON 3201
Econometrics for Economics and Finance

Assignment 1
Due Thursday October 8, 2026

Thiksha Sathish Kumar

Instructions

Answer all questions. Please complete your assignment using Quarto. Submit both your Quarto file and pdf output file.

Theory Questions

Q1

Create a personal Quarto website hosted on GitHup pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

i. An "Index" or "Home" page. This page should include the following:

- Your name
- Your photo (optional)
- Your credentials
- An "About Me" section that provides a brief (one paragraph) introduction to yourself. [10 Marks]

ii. A blog page. This will be the landing site for future assignments [5 Marks]
iii. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
iv. Navigation bar with "Home" and "Blog" buttons. The navigation bar is edited in the "_quarto.yml" file. [2 Marks]

> https://thiksha12.github.io/Thiksha-s-Website/

Q2

Write your solutions using LaTeX. Show your work. You may use R to compute the answers.

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Check that $f_X(x)$ is a true probability distribution.

The distribution is valid because all probabilities are non-negative and the probabilities add up to 1.

$$f_X(x) \geq 0$$

and

$$\sum_{x=1} f_X(x) = 1.$$

Since

$$f_X(x) = \frac{1}{12} > 0$$

for all $x = 1, 2, \ldots, 12$, and

$$\sum_{x=1}^{12} f_X(x) = \sum_{x=1}^{12} \frac{1}{12} = 12\left(\frac{1}{12}\right) = 1,$$

$f_X(x)$ is a true probability distribution.

(b) Compute the expected value of X.

$$E[X] = \sum_{x=1}^{12} x f_X(x).$$

Therefore,

$$\begin{aligned}
E[X] &= \sum_{x=1}^{12} x\left(\frac{1}{12}\right) \\
&= \frac{1}{12}\sum_{x=1}^{12} x \\
&= \frac{1}{12}(1+2+3+4+5+6+7+8+9+10+11+12) \\
&= \frac{1}{12}(78) \\
&= 6.5.
\end{aligned}$$

(c) Compute the variance of X.

The variance of X is

$$Var(X) = E[X^2] - [E[X]]^2.$$

First, compute $E[X^2]$:

$$\begin{aligned}
E[X^2] &= \sum_{x=1}^{12} x^2 f_X(x) \\
&= \sum_{x=1}^{12} x^2\left(\frac{1}{12}\right) \\
&= \frac{1}{12}(1^2 + 2^2 + \cdots + 12^2) \\
&= \frac{1}{12}(650) \\
&= \frac{325}{6}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
Var(X) &= E[X^2] - [E[X]]^2 \\
&= \frac{325}{6} - (6.5)^2 \\
&= \frac{325}{6} - 42.25 \\
&= 11.9167.
\end{aligned}$$

Thus,

$$Var(X) \approx 11.9167.$$

Q3

Let $X \sim N(4, 5)$. Find the probability of each event:

(a) $Pr(X \leq 6)$

$$Z = \frac{X - \mu}{\sigma}.$$

Since $\mu = 4$ and $\sigma = \sqrt{5}$,

$$P(X \leq 6) = P\left(Z \leq \frac{6-4}{\sqrt{5}}\right) = P(Z \leq 0.8944).$$

Using the standard normal distribution,

$$P(Z \leq 0.8944) \approx 0.8145.$$

Therefore,

$$P(X \leq 6) \approx 0.8145.$$

(b) $Pr(X > 4)$

$$P(X > 4) = P\left(Z > \frac{4-4}{\sqrt{5}}\right) = P(Z > 0).$$

By symmetry of the standard normal distribution,

$$P(Z > 0) = 0.5$$

Therefore,

$$P(X > 4) = 0.5$$

(c) $Pr(-1 \leq X < 1)$

$$\begin{aligned} P(-1 \leq X < 1) &= P\left(\frac{-1-4}{\sqrt{5}} \leq Z < \frac{1-4}{\sqrt{5}}\right) \\ &= P(-2.2361 \leq Z < -1.3416) \end{aligned}$$

Therefore,

$$\begin{aligned} P(-1 \leq X < 1) &= P(Z < -1.3416) - P(Z < -2.2361) \\ &\approx 0.0899 - 0.0127 \qquad = 0.0772. \end{aligned}$$

Therefore,

$$P(-1 \leq X < 1) \approx 0.0772.$$

Q4

Let $Z \sim N(0, 1)$.

(a) Compute $Pr(Z \leq 0.83)$

$$P(Z \leq 0.83) = 0.7967.$$

Therefore,

$$P(Z \leq 0.83) = 0.7967.$$

(b) Compute $Pr(-0.83 \leq Z \leq 0.83)$

Using the symmetry of the standard normal distribution,

$$P(-0.83 \leq Z \leq 0.83) = P(Z \leq 0.83) - P(Z < -0.83).$$

From the standard normal distribution table,

$$P(Z \leq 0.83) = 0.7967$$

and

$$P(Z < -0.83) = 0.2033.$$

Therefore,

$$\begin{aligned} P(-0.83 \leq Z \leq 0.83) &= 0.7967 - 0.2033 \\ &= 0.5934. \end{aligned}$$

Therefore,

$$P(-0.83 \leq Z \leq 0.83) = 0.5934$$

R Excercises

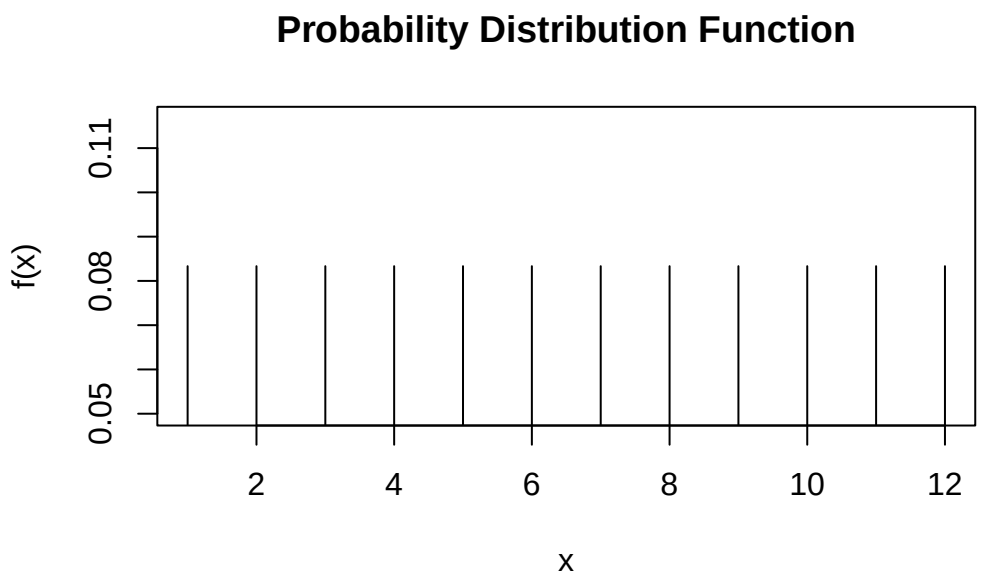
Q5

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Plot the probability distribution function $f_X(x)$ in R.

```
x= 1:12
p= rep(1/12, 12)

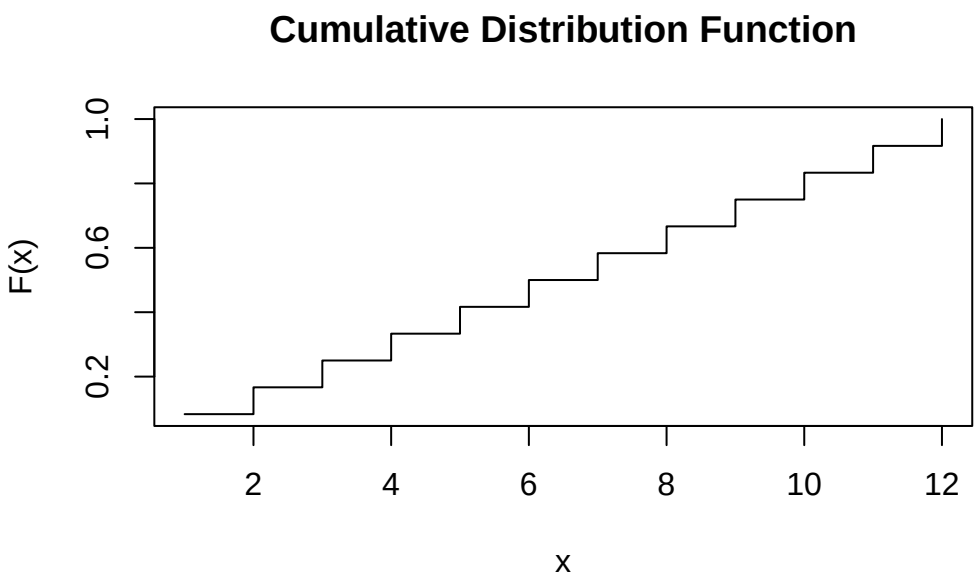
plot(x, p, type = "h", xlab = "x", ylab = "f(x)", main = "Probability Distribution Functio
```



(b) Plot the cumulative probability distribution function $f_X(x) in R.$

```
cdf= cumsum(p)

plot(x, cdf, type = "s", xlab = "x", ylab = "F(x)", main = "Cumulative Distribution Functi
```



(c) Compute the expected value of X in R.

```
expected_value= sum(x * p)
expected_value
```

```
[1] 6.5
```

(d) Compute the variance of X in R.

```
variance= sum((x - expected_value)^2 * p)
variance
```

```
[1] 11.91667
```

Q6

The standard normal distribution function is given by

$$f(x) = \frac{e^{-(x^2/2)}}{\sqrt{2\pi}\sigma}$$

(a) Define this function in R. (Note: π is coded as `pi` and e is coded as `exp`.

```
f= function(x) {(1 / sqrt(2 * pi)) * exp(-x^2 / 2)}
f
```

```
function (x)
{
    (1/sqrt(2 * pi)) * exp(-x^2/2)
}
```

(b) Integrate this function over the limit $[-\infty, \infty]$ using the `integrate()` command in R. Infinity in R is written as `Inf`. To evaluate the integral, use `integrate()$value`

```
total_prob= integrate(f, lower = -Inf, upper = Inf)$value
total_prob
```

```
[1] 1
```

(c) Compute the expected value of the standard normal distribution in R and confirm it is equal to 0.

```
expected_val_func= function(x) {x * f(x)}

expected_value= integrate(expected_val_func, lower = -Inf, upper = Inf)$value
expected_value
```

```
[1] 0
```

(d) Compute the variance of the standard normal distribution in R and confirm it is equal to 1.

```
variance_func= function(x) {(x^2) * f(x)}

variance= integrate(variance_func, lower = -Inf, upper = Inf)$value
variance
```

```
[1] 1
```

(e) Compute $Pr(Z \leq 0.83)$

```
prob_e= integrate(f, lower = -Inf, upper = 0.83)$value
prob_e
```

```
[1] 0.7967306
```

(f) Compute $Pr(-0.83 \leq Z \leq 0.83)$

```
prob_f= integrate(f, lower = -0.83, upper = 0.83)$value
prob_f
```

```
[1] 0.5934612
```

(g) Compute $Pr(-0.83 \leq Z \leq 0.83)$ using the `pnorm()` command and compare your results to (f).

```
prob_g= pnorm(0.83) - pnorm(-0.83)
prob_g
```

```
[1] 0.5934612
```

Q7

Let X denote the prison sentence, in years, for people convicted of auto theft in a particular province in Canada. Suppose the PDF of X is given by:

$$f(X) = \frac{1}{9}x^2, \ 0 < x < 3.$$

(a) Define $f(x)$ in R.

```
f= function(x) {(1 / 9) * x^2}
f
```

```
function (x)
{
    (1/9) * x^2
}
```

(b) Integrate $f(x)$ over the interval [0,3].

```
total_prob= integrate(f, lower = 0, upper = 3)$value
total_prob
```

```
[1] 1
```

(c) Compute the expected prison sentence.

The expected value is given by $E[X] = \int_0^3 x \cdot f(x)\, dx$:

```
# Function for x * f(x)
exp_func= function(x) {x * f(x)}

# Integrate x * f(x) from 0 to 3
expected_sentence= integrate(exp_func, lower = 0, upper = 3)$value
expected_sentence
```

```
[1] 2.25
```

Q8

Let X follow a Student t distribution with M degrees of freedom. Then

$$X = \frac{Z}{\sqrt{W/M}} \sim t_M,$$

where Z is a standard normal variable and W is a chi-squared variable with M degrees of freedom.

(a) On the same figure plot the following:

(i) Plot the standard normal probability distribution using `curve()`. Specify the range as [-5,5] using `xlim`. Name the `ylab` "Density". The main title should be "Densities of t Distribution".

(ii) Plot the Student t distribution with 5 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=2`. To learn more about `dt`, type `?dt` in your console.

(iii) Plot the Student t distribution with 15 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=3`. To learn more about `dt`, type `?dt` in your console.

(iv) Plot the Student t distribution with 25 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=4`. To learn more about `dt`, type `?dt` in your console.

(v) Include the following legend in you plot:
`legend("topright", c("N(0, 1)", "M=2", "M=4", "M=25"), col = 1:4, lty = c(1, 1, 1, 1))`

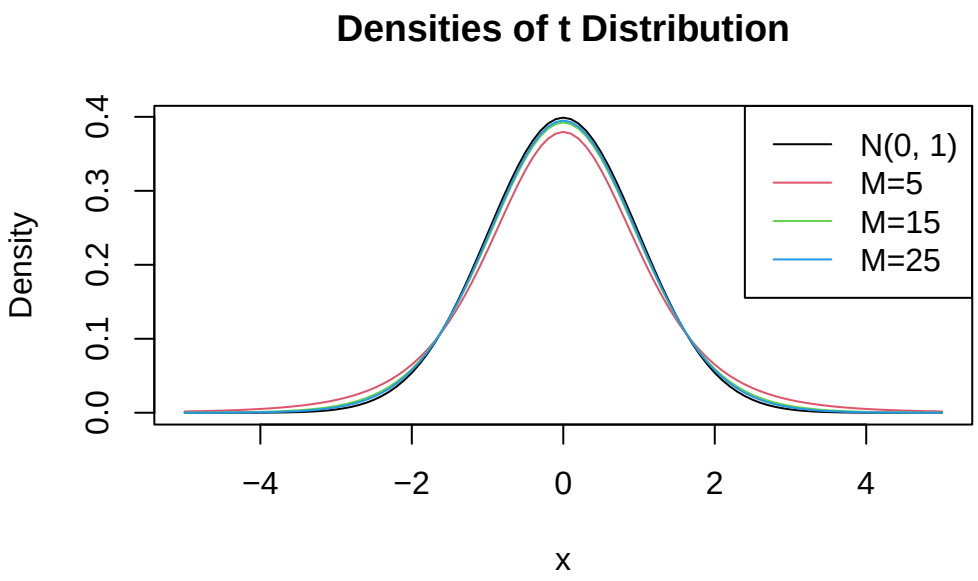
```r
# (i) Primary plot command - opens the plot window
curve(dnorm(x), xlim = c(-5, 5), ylab = "Density", main = "Densities of t Distribution", co

# (ii) Add t-distribution df = 5
curve(dt(x, df = 5), xlim = c(-5, 5), col = 2, add = TRUE)

# (iii) Add t-distribution df = 15
curve(dt(x, df = 15), xlim = c(-5, 5), col = 3, add = TRUE)

# (iv) Add t-distribution df = 25
curve(dt(x, df = 25), xlim = c(-5, 5), col = 4, add = TRUE)

# (v) Add legend
legend("topright", legend = c("N(0, 1)", "M=5", "M=15", "M=25"), col = 1:4, lty = 1)
```



(b) What do you observe as the degrees of freedom increase?

As the degrees of freedom (M) increase, the Student t-distribution approaches the standard normal distribution $N(0,1)$. Specifically, the tails become thinner, the peak becomes taller, and by $M = 25$, the t-distribution curve is identical to the standard normal curve.